DEPARTMENT OF COMPUTER APPLICATIONS
PSG COLLEGE OF TECHNOLOGY
MCA G1 & G2, SEMESTER 1
26MX16 - PROGRAMMING AND DATA STRUCTURES LABORATORY

PRACTICE SHEET 5

Prerequisite	2D traversal (Integer and String), Row traversal, Column traversal, Nested traversal, searching, position tracking, indexing and matrix transformation

1. Data Center Server Load Matrix

A data center records the CPU usage of R servers across C time intervals. Store the values in a 2D array and calculate the total CPU usage of each server.

Input Format

- First line: R C
- Next R lines: C integer CPU values

Sample Input

3 4
40 50 60 70
80 75 65 70
30 45 50 55

Sample Output

Server 1: 220
Server 2: 290
Server 3: 180

2. IT Helpdesk Ticket Matrix

An IT organization stores the number of support tickets received by different departments over several days. Find the department with the highest total number of tickets**.**

Sample Input

3 5
10 15 12 20 18
25 20 30 15 10
5 8 10 12 15

Sample Output

Department 2
Total Tickets: 100

3. Server Failure Locator

A monitoring system stores server status using:
1 → Working
0 → Failed
Find the number of failed servers in each time interval**.**

Sample Input

4 5
1 1 0 1 0
1 0 0 1 1
1 1 1 1 0
0 1 0 1 1

Sample Output

Time 1: 1 failure
Time 2: 2 failures
Time 3: 3 failures
Time 4: 0 failures
Time 5: 2 failures

4. Search an Employee ID

An organization stores employee IDs in a 2D array representing different teams. Given an employee ID, find its team number and position**.**

Sample Input

3 4

101 102 103 104
201 202 203 204
301 302 303 304
203

Sample Output

Employee Found
Team: 2
Position: 3

5. **Rotate a Matrix**

A cloud application stores server allocation data in a square matrix. Rotate the matrix 90 degrees clockwise**.**

Sample Input

3
1 2 3
4 5 6
7 8 9

Sample Output

7 4 1
8 5 2
9 6 3

6. **Find the Highest-Load Server**

A cloud platform stores CPU utilization of servers in a matrix.
Find:

- Highest CPU value
- Server number
- Time interval

Sample Input

3 4
45 60 55 70
80 75 65 90
50 55 85 60

Sample Output

Highest CPU Usage: 90
Server: 2
Time Interval: 4

7. **Product Inventory Matrix**

An e-commerce company stores product quantities in a matrix where each row represents a warehouse and each column represents a product. Find the total quantity of each product across all warehouses**.**

Sample Input

3 4
20 30 40 50
15 25 35 45
10 20 30 40

Sample Output

Product 1: 45
Product 2: 75
Product 3: 105
Product 4: 135

8. **Application Permission Matrix**

An enterprise application maintains a permission matrix:

1 → Access Granted
0 → Access Denied
Rows represent employees and columns represent applications.
Given an employee number, count the number of applications to which the employee has access.

Sample Input
4 5
1 1 0 1 0
1 0 1 1 1
0 0 1 0 1
1 1 1 1 1
2
Sample Output
Applications Accessible: 4

9. Word Search in an Employee Directory ⋆

An organization stores employee names in a 2D string array. Given a search string, determine whether the name exists. The search must be case-sensitive.

Sample Input
3 3
Arun Bala Charan
Divya Ezhil Farah
Gokul Hari Isha
Farah
Sample Output
FOUND
Row: 2
Column: 3
Sample Input 2
3 3
Arun Bala Charan
Divya Ezhil Farah
Gokul Hari Isha
Rahul
Sample Output 2
NOT FOUND

10. Find Duplicate Usernames ⋆

A software application stores usernames in a 2D array representing different servers. Find whether any username appears more than once.

Sample Input
3 3
admin user1 user2
user3 admin user4
user5 user6 user2
Sample Output
Duplicate Usernames:
admin
user2

11. Diagonal Server Analysis ⋆⋆

A monitoring system stores server performance scores in a square matrix.
Calculate:

- Sum of the main diagonal

- Sum of the secondary diagonal
- Difference between the two sums

Sample Input

4
10 20 30 40
50 60 70 80
90 15 25 35
45 55 65 75

Sample Output

Main Diagonal Sum: 170
Secondary Diagonal Sum: 160
Difference: 10

12. IT Department Performance Analyzer ⋆⋆

An organization records employee performance scores using a 2D array. Each row represents a department and each column represents an employee.
For every department:

- Find the highest score.
- Find the lowest score.
- Calculate the number of employees scoring above 80.

Sample Input

3 4
75 85 90 65
88 72 95 91
60 70 82 85

Sample Output

Department 1
Highest: 90
Lowest: 65
Above 80: 2

Department 2
Highest: 95
Lowest: 72
Above 80: 3

Department 3
Highest: 85
Lowest: 60
Above 80: 2

13. Smart IT Service Dashboard ⋆⋆⋆

An IT service platform maintains a matrix containing the number of incidents handled by different support teams across several days. A high-load day is a day where the total number of incidents is greater than 100.
Determine:

- Total incidents handled by each team.
- The day with the highest number of incidents.
- Number of high-load days.
- The team with the highest total workload.

Sample Input

3 5
20 25 30 35 40
15 20 25 30 35

30 25 20 35 40

Sample Output

Team 1 Total: 150
Team 2 Total: 125
Team 3 Total: 150

Peak Day: 5
Peak Day Incidents: 115

High-Load Days: 2

Highest Workload Team: 1

Skills Acquired	2D Array Traversal, Row/Column Processing, Nested Iteration, Integer and String Searching, Index and Position Tracking and Matrix Transformation.